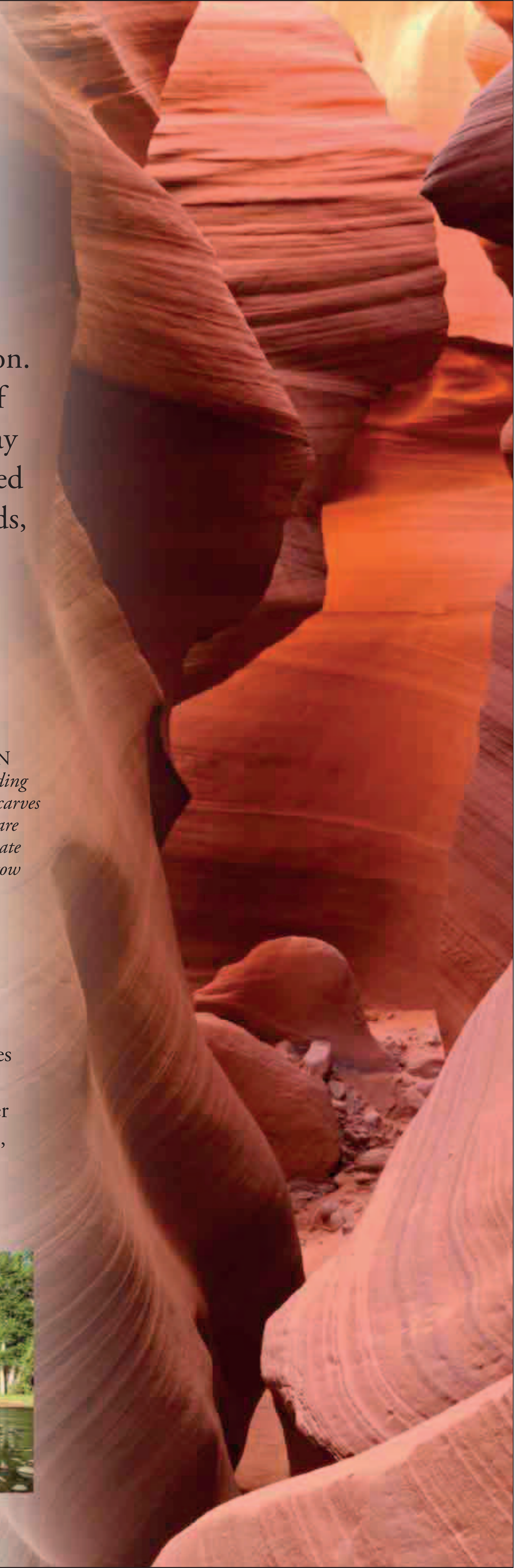
The dynamic landscape

Rock that has been broken down by weathering is worn away by other forces in a process called erosion. A lump of granite may crumble under the attack of acidified rainwater, but it is erosion that scours away the resulting clay and sand particles. They are carried off by flowing water, moving ice, or even high winds, and deposited somewhere else as various types of sediment. The process of erosion is constantly reshaping the landscape.

SCOURED BY WATER

Most of the particles that crumble from weathered rocks are carried away by flowing water. When this flows over limestone, it gradually dissolves the rock, but usually the erosion is not caused by the water itself. It is caused by hard mineral particles that are suspended in the water. They scour rock faces like sandpaper and carve them away, especially when the water is flowing fast.



▸ SLOT CANYON
Flash flooding in deserts carves through bare rock to create deep, narrow canyons.

SHINGLE, SAND, AND SILT

Rivers carry small particles suspended in the water, and this often makes them look muddy. Their flow bumps bigger, heavier stones along the riverbed, rounding off their sharp corners to form shingle. When a river reaches flatter ground, it slows down and drops first the heaviest stones, then the smaller ones. It carries the small sand grains farther, and may carry the tiniest silt and clay particles all the way to the sea.